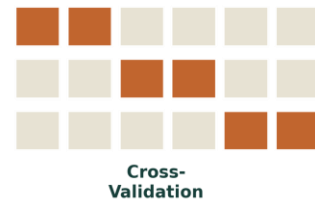
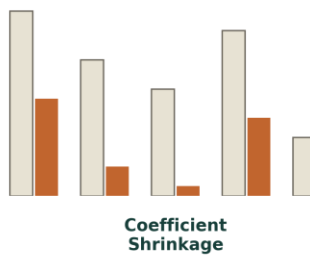
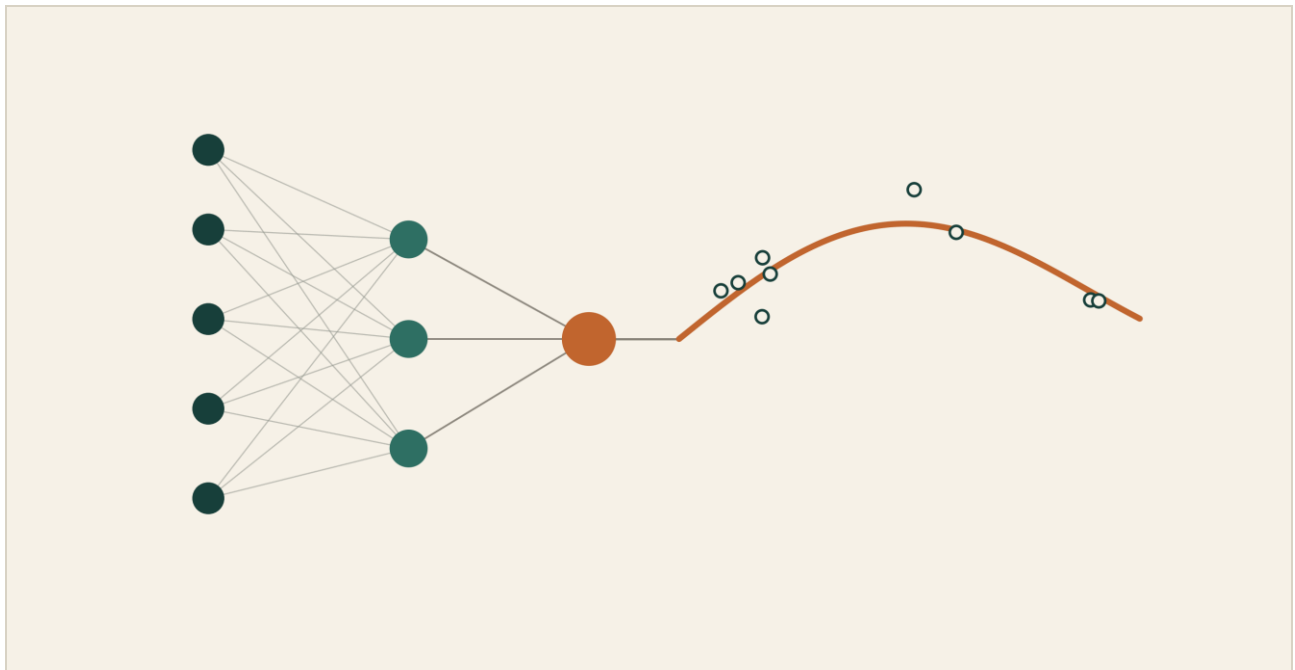


ROBUST REGRESSION ENGINE

Part :- A Conceptual Foundation (Theory)



🌀 Part A: Conceptual Foundation (Theory)

Answer and supported with diagrams for visual clarity.

1 What is Regularization in Machine Learning? Why is it needed?

Regularization adds a penalty term to a model's loss function that discourages excessively large coefficient values. It is needed because unconstrained models tend to fit noise in the training data (overfitting), which hurts performance on unseen data. By shrinking coefficients, regularization trades a small increase in bias for a large reduction in variance, producing a model that generalizes better.



Figure 1 — An unregularized model chases every data point, while a regularized model favors a smoother, more general fit.

2 Difference between Ridge Regression (L2) and Lasso Regression (L1)

Both add a penalty proportional to the size of the coefficients, but differ in form and effect. Ridge (L2) penalizes the sum of squared coefficients, shrinking them smoothly toward zero without eliminating any — useful when most features are relevant. Lasso (L1) penalizes the sum of absolute coefficients, which can shrink some coefficients exactly to zero, performing automatic feature selection.

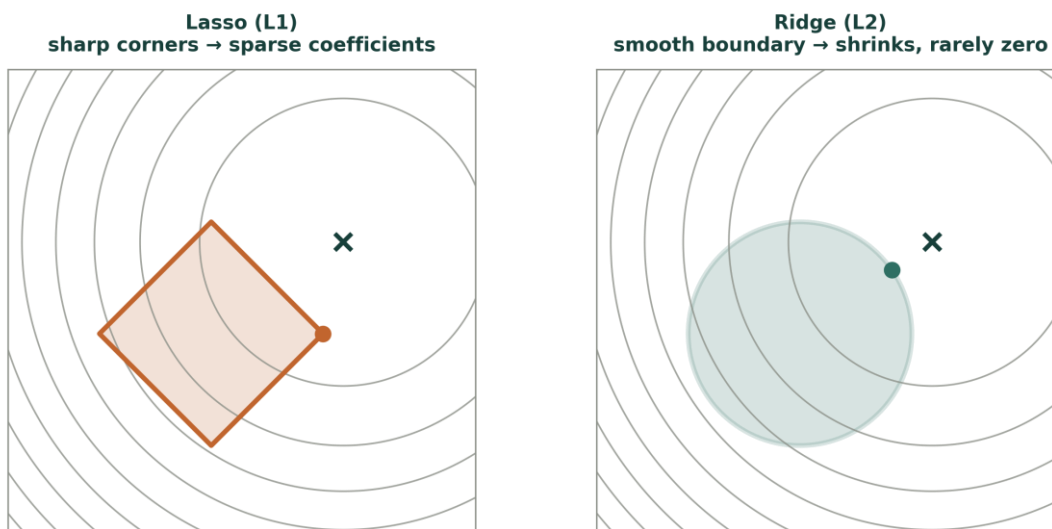


Figure 2 — The diamond-shaped L1 constraint region meets the error contours at a corner (a zeroed coefficient); the circular L2 region shrinks coefficients smoothly instead.

3 What is Cross-Validation and why is it important?

Cross-validation is a resampling technique that repeatedly splits data into training and validation subsets to evaluate how a model performs on data it has not seen. It is important because it gives a more reliable estimate of generalization performance than a single train/test split, and helps detect overfitting before deployment.

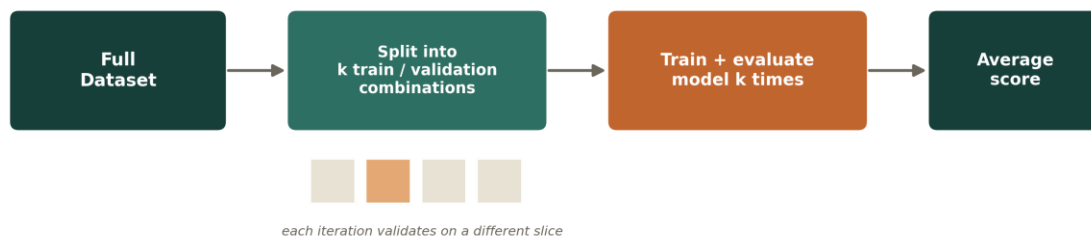


Figure 3 — Cross-validation repeats training and evaluation across several data slices, then averages the results into one performance estimate.

4 Explain the following cross-validation techniques

- K-Fold Cross-Validation — the data is split into k equal folds; the model trains on $k-1$ folds and validates on the remaining fold, rotating until every fold has served as the validation set once.
- Stratified K-Fold Cross-Validation — like K-Fold, but each fold preserves the same class proportions as the full dataset, which matters for imbalanced classification problems.
- Leave-One-Out Cross-Validation (LOOCV) — the extreme case of K-Fold where k equals the number of samples, so each iteration validates on a single data point.
- Time Series Split — folds respect chronological order so training data always precedes the validation window, preventing the model from "seeing the future."



Figure 4 — Copper cells mark the validation portion in each iteration for all four cross-validation strategies.

5 Why are tree-based models less sensitive to feature scaling?

Tree-based models (Decision Trees, Random Forests, Gradient Boosting) split nodes by comparing a feature to a threshold rather than computing distances or weighted sums across features. Since any monotonic transformation — including scaling — preserves the relative ordering of values, the chosen split points and resulting tree structure remain unchanged.

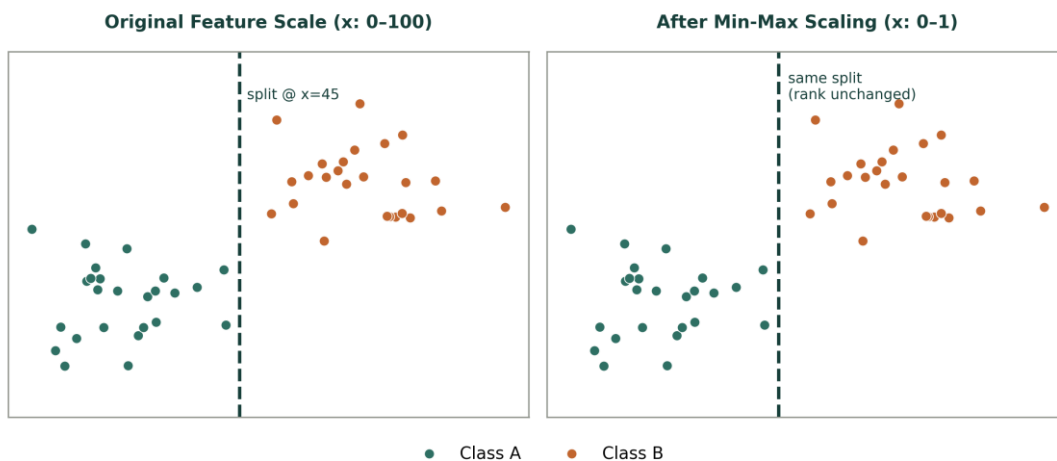


Figure 5 — The decision split lands on the same data points before and after Min-Max scaling, because the split depends only on rank order.